Technical Report: Digital Heritage - Great Chicago Fire 1871

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Abstract

This project represents the transformation of historical material about the Great Chicago Fire of 1871 into an interactive digital space demonstrating collective memory formation through media. The project is implemented as a one-page multimedia website using HTML5, CSS3, and JavaScript. The main goal of the work is to show how digital tools can be used to preserve and interpret cultural memory of urban disasters, transforming historical material into an interactive educational space.

1. Introduction

1.1 Research Relevance

The Great Chicago Fire of 1871 represents a unique case for studying collective memory formation through media. This event not only became the largest disaster in 19th century US history, but also gave rise to one of the most famous urban myths - the legend of Mrs. O'Leary's cow.

Modern approaches in Digital Heritage allow the use of digital technologies to preserve and interpret historical memory. Our project demonstrates how interactive tools can transform traditional historical sources into new types of educational spaces.

1.2 Research Goal and Objectives

Research Goal:

Transformation of historical material about the Great Chicago Fire of 1871 into an interactive digital space demonstrating collective memory formation through media.

Research Objectives:

1. Analyze historical sources (Harper's Weekly illustrations, 1872 maps)
2. Investigate the process of "O'Leary's cow" myth formation
3. Create a one-page multimedia website
4. Implement interactive elements (slider, animations, game, quiz)
5. Ensure multilingualism (Russian, English, Italian, Turkish)
6. Demonstrate Digital Heritage approach

2. Theoretical Framework

2.1 Digital Heritage

Digital Heritage is a field concerned with using digital technologies to preserve, document, and interpret cultural heritage. In the context of our project, the Digital Heritage approach means:

- Preserving historical sources in digital format
- Creating interactive mechanisms for accessing historical information
- Using modern technologies for new interpretation of historical events
- Ensuring educational value through interactivity

2.2 Collective Memory

The concept of collective memory, developed by Maurice Halbwachs, views memory as a social construct. In our project, we study:

- How media (Harper's Weekly, 1871) shaped the image of disaster
- The process of myth construction (O'Leary cow legend, 1893)
- The evolution of city image from "ruin cult" to industrial hope
- The role of modern digital technologies in representing historical memory

3. Methodology

3.1 Sources

The project is based on the following primary sources:

- **Harper's Weekly -** illustrations from October-November 1871
- **Map of Chicago's Burnt District -** 1872
- **Chicago Historical Society -** archival materials
- **Encyclopedia of Chicago -** event chronology

3.2 Development Technologies

HTML5	3556 lines - website structure
CSS3	~800 lines - styling and animations
JavaScript	~1200 lines - interactivity
Canvas API	3 animation layers (flames, sparks, ash)

4. Results

4.1 Created Components

- **One-page website -** 10 sections with various content
- **Interactive slider -** comparing "press vs map"
- **Canvas animations -** atmospheric effects (flames, sparks, ash)
- **Interactive game -** "Save Chicago" with Canvas gameplay
- **Quiz -** 30 questions about the fire
- **Multilingualism -** support for 4 languages (RU, EN, IT, TR)

4.2 Project Statistics

Total code volume	5556 lines
Sections	10
Supported languages	4
Quiz questions	30
Canvas animations	3
Video materials	7+

5. Conclusions

The project successfully demonstrates how digital tools can transform historical material into an interactive educational space. Main conclusions:

- Interactive comparative approach allows clear demonstration of myth formation process
- Canvas API enables creation of atmospheric visual effects immersing users in historical era
- Multilingualism ensures accessibility for international audience
- Historical accuracy achieved through use of primary sources
- Digital Heritage approaches enable creation of new types of educational tools

Practical Significance

The project can be used in educational purposes: in US history classes, Digital Humanities courses, for studying collective memory formation, as an example of interactive storytelling.